



HAWC Hackathon 2026

Over a century of weather in Reading

Atmospheric Observatory, University of Reading | 2026

Event Kick-off Session	9:00am – 9:30am, Wednesday 2 September 2026
Presentations & Awards	2:00pm – 4:00pm, Thursday 3 September 2026
Venue	Room 1L61, Department of Meteorology, Brian Hoskins Building, University of Reading, Whiteknights Campus

Contents

1	About the HAWC Hackathon 2026	1
2	Themes	1
3	Judging criteria	2
4	Schedule	2
5	What to submit	2
6	Data & links	3
	Essential links	3
6.1	Datasets	3
6.2	Other resources	3

1 About the HAWC Hackathon 2026

The HAWC Hackathon 2026 challenges participants to make creative use of one of Reading’s longest-running scientific records. Since 1901, the University of Reading has collected near-continuous weather observations, first at its London Road campus and, since 1968, at the Atmospheric Observatory on the Whiteknights campus. Observations have been made by hand almost every day, by generations of observers recording the same variables consistently over more than a century. With this data, you’ll have the opportunity to:

- Create tools that help communities prepare for weather and climate change.
- Inspire awareness and engagement by presenting data in fresh, accessible ways.
- Build prototypes that could shape research, education, and policy decisions.

2 Themes

Choose one theme to focus your project on. Judges will evaluate submissions against the theme you enter.

Theme 1 — Hyperlocal forecasting tools

Build tools that support smarter decision-making for communities, businesses and local authorities. Judges will be looking for interesting approaches to forecasting meteorological variables and innovative uses of the observatory’s data.

Theme 2 — Weather apps, webpages and visualisations

Make Reading’s weather history accessible, engaging and useful to the public. Judges will be looking for visually appealing outputs that tell a story, strong interactivity and a clear focus on weather and climate variables.

Theme 3 — Weather connects to everything

Link Reading’s unique weather record to data from other fields, such as energy demand, agriculture or human geography. Judges will be looking for interesting connections between weather data and non-weather data local to Reading, and a compelling data story that brings them together.

3 Judging criteria

Projects will be assessed by a panel of judges against the following criteria:

Criterion	What judges are looking for
Originality	A fresh idea, approach or angle – not just a rehash of an existing tool or dataset view.
Significance & potential impact	Real-world relevance: does the project matter to communities, businesses, researchers or policymakers in and around Reading?
Technical execution	Quality and appropriateness of the methods used, and how well the prototype actually works.
Use of data	Effective, thoughtful and (where relevant) innovative use of the Observatory's data, alone or combined with other sources.
Communication & presentation	How clearly and compellingly the team explains the problem, approach and results in their presentation.

4 Schedule

When	What
9:00am – 9:30am, Wednesday 2 September 2026	Opening session. Welcome talk(s), team formation (for those who need it), and hacking start!
<i>Mentor / check-in slots</i>	Ieuan Higgs will be available throughout the event to discuss ideas, offer advice and give feedback on WIPs.
1:00pm, Thursday 3 September	Submissions close. Projects must be submitted by this time. See Section 5 for what to submit.
2:00pm – 4:00pm, Thursday 3 September 2026	Presentations and awards. Teams present their work to judges, followed by the announcement of winners.

5 What to submit

By 1:00pm, Thursday 3 September, each team should submit their projects to <https://forms.cloud.microsoft/e/KKQVi95Xjb>. Be sure to prepare:

- **Code and a short report** – a link to your public GitHub/GitLab repository, containing your code and, ideally, your summary write-up alongside it. The report itself can be as simple as a one-page PDF or README covering the problem you tackled, your approach, and what you built. Remember to state which of the three themes (Section 2) your project addresses, and which dataset(s) you used.
- **A presentation** – each team will also present their work, with approximately 10-15 minutes during the presentation session to show off what you've created. This could be a PowerPoint, a live demo, key figures, or a combination of the three, whatever best lets you present your project.

6 Data & links

Essential links

Submit your project <https://forms.cloud.microsoft/e/KKQVi95Xjb>

Questions? Dr. Ieuan Higgs: i.c.higgs@reading.ac.uk

6.1 Datasets

- **Reading manual weather observations (1901–present)** – the University of Reading’s long-run manual climatological record, curated by Dr. Stephen Burt. Daily observations of temperature, rainfall, sunshine, pressure, wind, cloud and other elements have been taken once each day at 0900 UTC since 1901, with an almost complete series from January 1908 onward; the station moved from the London Road campus to the Atmospheric Observatory at Whiteknights in 1968, and manual observations continue to be made 365 days a year. Dataset upload: https://github.com/ichiggs/HAWC_hackathon_2026.
- **Reading Atmospheric Observatory automatic measurements (1997–2023)** – 5-minute surface meteorological observations (temperature, rainfall, wind, pressure, humidity and more) in CSV format: <https://doi.org/10.17864/1947.000490>
- **HAWC Open Data Catalogue** – browse all datasets curated by the Hub for Applied Weather and Climate Research, including the observatory record and related climate/energy datasets: <https://research.reading.ac.uk/applied-weather-climate/open-data-catalogue/>
- **CEDA catalogue record** – platform metadata for the Observatory: <https://catalogue.ceda.ac.uk/uuid/dad6ee58123d405cb3d7afa5f3d29d7d/>
- **Ogimet** – free access to historical SYNOP, METAR and TAF weather reports from thousands of stations worldwide, useful for comparing Reading’s record against other locations: <https://www.ogimet.com/home.phtml.en>
- **You are welcome to find your own additional datasets.** e.g., local energy demand, transport, agriculture or population data. You can be very creative with the the “Weather connects to everything” theme!

6.2 Other resources

- **HAWC – Hub for Applied Weather and Climate Research:** <https://research.reading.ac.uk/applied-weather-climate/>
- **About the Atmospheric Observatory** – background on the site, instruments and the 100+ year observational record: <https://research.reading.ac.uk/meteorology/atmospheric-observatory/>